Exercise 8 (solved in class together)

$$U(x-\Delta x) = U(x) - \left(\frac{\partial U}{\partial x}\right)\Delta x + \mathcal{O}(\Delta x^2)$$

$$\underline{U(x-\Delta x)} = U(x) - \frac{\left(\frac{\partial U}{\partial x}\right)\cancel{\Delta x} + \mathcal{O}(\cancel{\Delta x}^2)}{\cancel{\Delta x}}$$

$$\boxed{\frac{\partial U}{\partial x} - \mathcal{O}(\Delta x)} = \frac{-[U(x-\Delta x) - U(x)]}{\Delta x}$$

$$\boxed{\frac{\partial U}{\partial x}} = -\frac{[U(x-\Delta x) - U(x)]}{\Delta x} + \mathcal{O}(\Delta x)$$

$$\boxed{\frac{\partial U}{\partial x}} \approx -\frac{[U_i - U_{i-1}]}{\Delta x}$$

- $x \rightarrow i$
- $x - \Delta x \rightarrow i-1$

Exercise 9

- $x_i = x_0 + i(\Delta x)$
- $t^n = t_0 + n(\Delta t)$

e 29

$$U(x+\Delta x) = U(x) + \left(\frac{\partial U}{\partial x}\right)\Delta x + \frac{1}{2}\left(\frac{\partial^2 U}{\partial x^2}\right)\Delta x^2 + O(\Delta x^3)$$

$$U(x-\Delta x) = U(x) - \left(\frac{\partial U}{\partial x}\right)\Delta x + \frac{1}{2}\left(\frac{\partial^2 U}{\partial x^2}\right)\Delta x^2 + O(\Delta x^3)$$

$$U(x+\Delta x) + U(x-\Delta x) = 2U(x) + \left(\frac{\partial^2 U}{\partial x^2}\right)\Delta x^2 + O(\Delta x^4)$$

$$\frac{\partial^2 U}{\partial x^2} = \frac{U(x+\Delta x) - 2U(x) + U(x-\Delta x)}{\Delta x^2} + O(\Delta x^2)$$

$$\frac{\partial^2 U}{\partial x^2} \approx \frac{U_{i+1} - 2U_i + U_{i-1}}{\Delta x^2}$$

I got help from Claude AI with equation 29, that I used to discretise equation 33, had minor sign mistakes that I fixed.

e 33

$$U(z+\Delta z) = U(z) + \left(\frac{\partial U}{\partial z}\right)\Delta z + \frac{1}{2}\cancel{\left(\frac{\partial^2 U}{\partial z^2}\right)}\Delta z^2 + O(\Delta z^3)$$

$$U(z-\Delta z) = U(z) - \left(\frac{\partial U}{\partial z}\right)\Delta z + \frac{1}{2}\cancel{\left(\frac{\partial^2 U}{\partial z^2}\right)}\Delta z^2 + O(\Delta z^3)$$

$$U(z+\Delta z) - U(z-\Delta z) = 2\left(\frac{\partial U}{\partial z}\right)\Delta z + O(\Delta z^3)$$

$$\left(\frac{\partial U}{\partial z}\right) = \frac{U(z+\Delta z) - U(z-\Delta z)}{2\Delta z} + O(\Delta z^2)$$

$$\frac{\partial U}{\partial z} = \frac{U_{k+1} - U_{k-1}}{2\Delta z}$$

$$\frac{\partial U}{\partial z} = \frac{U^n_{j,k+1} - U^n_{j,k-1}}{2\Delta z}$$